Lesson 10: Data Preparation and Transformation using Tableau Prep

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What you will learn ?
a) Connect and load data into Tableau Prep
b) Clean and profile data
c) Group and standardize values
d) Merge mismatched fields
e) Perform aggregations in Prep
f) Apply filters and transformations
g) Create and automate data preparation flows

You need Tableau Prep Builder for this Lesson. You need to download and install. Here are the steps :

1. Download
- Download the Windows installer (.exe) directly from the official **Tableau Prep Builder Releases Page** *(or use the **Tableau Prep Free Trial Page***).

2. Install
- Open your **Downloads** folder and run the .exe file.
- Accept the license agreement, click **Install**, and confirm **Yes** on the Windows prompt.

3. Activate & Launch
- Open **Tableau Prep Builder** from the Start Menu.
- Select **Activate with a product key** (enter your Tableau Desktop/Student key) or **Start a Free Trial**.

Dataset: Custom Designed Students Dataset [csv files]

You can also create database using Python code. Here is the code

```
import pandas as pd
import numpy as np

# We need custom CSV datasets specifically designed for Tableau
Prep features in Practical 10:
# a) Connect & load data
# b) Clean and profile data (data type issues, extra spaces,
nulls, outliers)
#  c)  Group  and  standardize  values  (pronunciation/spelling
variations, e.g. "Comp Sci", "CS", "Computer Science", "CSE",
"Electronic Eng", "ECE")
# d) Merge mismatched fields (e.g., across two unioned files
or separate columns: "Maths" vs "Mathematics", "Std_ID" vs
"Student_ID")
# e) Perform aggregations in Prep (e.g., Aggregate step: Total
Score, Subject Average, Group by Student/Term)
#  f)  Apply  filters  and  transformations  (e.g.,  Filter  out
dropped students, calculate weighted GPA / Grade, split
columns)
```

```
# g) Create and automate data preparation flows (Output step to .hyper / .csv)

# Table 1: Raw Student Assessment Log (Term 1) -> Student_Assessments_Term1.csv
# Contains messy department names for (c) Group & Standardize, inconsistent casings, text dates, extra spaces

term1_data = {
    'Student_ID': ['S101', 'S101', 'S102', 'S102', 'S103', 'S103', 'S104', 'S104', 'S105', 'S105', 'S106', 'S106', 'S107', 'S107', 'S108', 'S108'],
    'Student_Name': [' Aarav Sharma ', 'Aarav Sharma', 'priya VERMA', 'priya VERMA', 'Rohan Gupta', 'Rohan Gupta', 'Sneha Patel', 'Sneha Patel', 'aditya Rao ', 'aditya Rao ', 'Ananya Singh', 'Ananya Singh', 'Rahul Nair', 'Rahul Nair', 'Kavya Reddy', 'Kavya Reddy'],
    'Department': ['Computer Science', 'Comp Sci', 'Electronics', 'ECE', 'CS', 'CSE', 'Information Tech', 'IT', 'Mechanical', 'Mech Engg', 'Computer Science', 'CS', 'ECE', 'Electronics Engg', 'IT', 'Info Tech'],
    'Assessment_Type': ['Midterm', 'Final', 'Midterm', 'Final', 'Midterm', 'Final', 'Midterm', 'Final', 'Midterm', 'Final', 'Midterm', 'Final', 'Midterm', 'Final', 'Midterm', 'Final'],
    'Exam_Date': ['2025/09/15', '2025/11/20', '15-09-2025', '20-11-2025', '2025.09.15', '2025.11.20', '2025/09/15', '2025/11/20', '2025/09/15', '2025/11/20', '2025/09/15', '2025/11/20', '2025/09/15', '2025/11/20', '2025/09/15', '2025/11/20'],
    'Subject_Code': ['CS101', 'CS101', 'EC101', 'EC101', 'CS101', 'CS101', 'IT101', 'IT101', 'ME101', 'ME101', 'CS101', 'CS101', 'EC101', 'EC101', 'IT101', 'IT101'],
    'Marks_Obtained': [85, 92, 70, 78, 45, 52, 90, 96, 35, 40, 88, 94, 65, 72, 92, 98],
    'Max_Marks': [100, 100, 100, 100, 100, 100, 100, 100, 100, 100, 100, 100, 100, 100, 100, 100],
    'Enrollment_Status': ['Active', 'Active', 'Active', 'Active', 'Active', 'Active', 'Active', 'Active', 'Dropped', 'Dropped', 'Active', 'Active', 'Active', 'Active', 'Active', 'Active']
}

# Table 2: Student Assessment Log (Term 2) with Mismatched Column Headers -> Student_Assessments_Term2.csv
# For (d) Merge Mismatched Fields (e.g. Std_ID instead of Student_ID, Score instead of Marks_Obtained, Dept instead of Department)

term2_data = {
    'Std_ID': ['S101', 'S101', 'S102', 'S102', 'S103', 'S103', 'S104', 'S104', 'S106', 'S106', 'S107', 'S107', 'S108', 'S108'],
```

```
    'Full_Name': ['Aarav Sharma', 'Aarav Sharma', 'Priya
Verma', 'Priya Verma', 'Rohan Gupta', 'Rohan Gupta', 'Sneha
Patel', 'Sneha Patel', 'Ananya Singh', 'Ananya Singh', 'Rahul
Nair', 'Rahul Nair', 'Kavya Reddy', 'Kavya Reddy'],
    'Dept': ['Computer Science', 'CS', 'ECE', 'Electronics',
'CSE', 'CS', 'IT', 'Information Tech', 'CSE', 'Computer
Science', 'ECE', 'Electronics', 'IT', 'IT'],
    'Exam_Type': ['Midterm', 'Final', 'Midterm', 'Final',
'Midterm', 'Final', 'Midterm', 'Final', 'Midterm', 'Final',
'Midterm', 'Final', 'Midterm', 'Final'],
    'Assessment_Date': ['2026/02/10', '2026/04/25',
'2026/02/10', '2026/04/25', '2026/02/10', '2026/04/25',
'2026/02/10', '2026/04/25', '2026/02/10', '2026/04/25',
'2026/02/10', '2026/04/25', '2026/02/10', '2026/04/25'],
    'Course_Code': ['CS201', 'CS201', 'EC201', 'EC201',
'CS201', 'CS201', 'IT201', 'IT201', 'CS201', 'CS201', 'EC201',
'EC201', 'IT201', 'IT201'],
    'Score': [88, 95, 75, 82, 50, 60, 94, 98, 90, 96, 70, 75,
95, 100],
    'Total_Possible': [100, 100, 100, 100, 100, 100, 100, 100,
100, 100, 100, 100, 100, 100],
    'Enrollment_Status': ['Active', 'Active', 'Active',
'Active', 'Active', 'Active', 'Active', 'Active', 'Active',
'Active', 'Active', 'Active', 'Active', 'Active']
}

# Table 3: Course Credit & Grading Benchmark ->
Course_Credit_Master.csv

course_data = {
    'Subject_Code': ['CS101', 'CS201', 'EC101', 'EC201',
'IT101', 'IT201', 'ME101'],
    'Subject_Title': ['Data Structures', 'Algorithms',
'Digital Logic', 'Signals & Systems', 'Database Systems',
'Cloud Computing', 'Thermodynamics'],
    'Credit_Hours': [4, 4, 3, 4, 4, 4, 3],
    'Passing_Marks': [40, 40, 40, 40, 40, 40, 40]
}

df_t1 = pd.DataFrame(term1_data)
df_t2 = pd.DataFrame(term2_data)
df_c = pd.DataFrame(course_data)

df_t1.to_csv('Student_Assessments_Term1.csv', index=False)
df_t2.to_csv('Student_Assessments_Term2.csv', index=False)
df_c.to_csv('Course_Credit_Master.csv', index=False)

print("All 3 CSV datasets for Lesson 10 created successfully.")
```

Here is the complete schema design and CSV datasets tailored specifically for Tableau Prep Builder to fulfill all activities of Practical 10 (a through g).

Dataset Architecture and Schema Design

To practice every feature of Tableau Prep (profiling, grouping, merging mismatched union columns, aggregations, calculated filters, and automated flow generation), we use 3 specialized CSV files:

1. Student_Assessments_Term1.csv (Term 1 Assessment Log)
   - Purpose: Data profiling, handling extra spaces, mixed casings, inconsistent date text strings, and spelling variations in departments.
   - Columns: Student_ID, Student_Name, Department, Assessment_Type, Exam_Date, Subject_Code, Marks_Obtained, Max_Marks, Enrollment_Status.
2. Student_Assessments_Term2.csv (Term 2 Assessment Log with Mismatched Schemas)
   - Purpose: Practicing Union and resolving mismatched field names (e.g., Std_ID vs Student_ID, Score vs Marks_Obtained, Course_Code vs Subject_Code).
   - Columns: Std_ID, Full_Name, Dept, Exam_Type, Assessment_Date, Course_Code, Score, Total_Possible, Enrollment_Status.
3. Course_Credit_Master.csv (Subject Metadata & Weightages)
   - Purpose: Prep Joins, weighted GPA calculations, credit hours, and subject categorization.
   - Columns: Subject_Code, Subject_Title, Credit_Hours, Passing_Marks.

Mapping to Practical 10 Objectives

Sub-task	Activity	Practical Execution in Tableau Prep
a	Connect and load data	Connect to text files (Term1, Term2, and Course_Master) in Tableau Prep.
b	Clean and profile data	Use Prep's Profile pane to inspect value distributions, summary cards, and detect data type errors.
c	Group and standardize values	Use Group Values (Pronunciation, Common Characters, or Manual) to unify "Comp Sci", "CS", "CSE" --> "Computer Science".
d	Merge mismatched fields	Union Term 1 and Term 2, then drag and drop mismatched columns (Std_ID onto Student_ID, Score onto Marks_Obtained) to merge them.
e	Perform aggregations in Prep	Add an Aggregate Step to compute student total score, overall average, and exam count grouped by Student_ID & Term.

Sub-task	Activity	Practical Execution in Tableau Prep
f	Apply filters and transformations	Filter out "Dropped" enrollment records, create calculated fields for Percentage, Grade, and Status flags.
g	Create and automate data flows	Build an Output Step to export the cleaned, unified pipeline to a .hyper extract or .csv file.

Raw CSV Data (Create these 3 files on your computer)**.** You can create different csv files using Notepad/text Editor. Copy paste the data in Notepad/text Editor and save the files with .csv extension.

File 1: Student_Assessments_Term1.csv

```
Student_ID,Student_Name,Department,Assessment_Type,Exam_Date,
Subject_Code,Marks_Obtained,Max_Marks,Enrollment_Status
S101,  Aarav Sharma ,Computer
Science,Midterm,2025/09/15,CS101,85,100,Active
S101,Aarav Sharma,Comp
Sci,Final,2025/11/20,CS101,92,100,Active
S102,priya VERMA,Electronics,Midterm,15-09-
2025,EC101,70,100,Active
S102,priya VERMA,ECE,Final,20-11-2025,EC101,78,100,Active
S103,Rohan Gupta,CS,Midterm,2025.09.15,CS101,45,100,Active
S103,Rohan Gupta,CSE,Final,2025.11.20,CS101,52,100,Active
S104,Sneha Patel,Information
Tech,Midterm,2025/09/15,IT101,90,100,Active
S104,Sneha Patel,IT,Final,2025/11/20,IT101,96,100,Active
S105,aditya Rao
,Mechanical,Midterm,2025/09/15,ME101,35,100,Dropped
S105,aditya Rao ,Mech
Engg,Final,2025/11/20,ME101,40,100,Dropped
S106,Ananya Singh,Computer
Science,Midterm,2025/09/15,CS101,88,100,Active
S106,Ananya Singh,CS,Final,2025/11/20,CS101,94,100,Active
S107,Rahul Nair,ECE,Midterm,2025/09/15,EC101,65,100,Active
S107,Rahul Nair,Electronics
Engg,Final,2025/11/20,EC101,72,100,Active
S108,Kavya Reddy,IT,Midterm,2025/09/15,IT101,92,100,Active
S108,Kavya Reddy,Info
Tech,Final,2025/11/20,IT101,98,100,Active
```

File 2: Student_Assessments_Term2.csv

```
Std_ID,Full_Name,Dept,Exam_Type,Assessment_Date,Course_Code,S
core,Total_Possible,Enrollment_Status
S101,Aarav Sharma,Computer
Science,Midterm,2026/02/10,CS201,88,100,Active
S101,Aarav Sharma,CS,Final,2026/04/25,CS201,95,100,Active
S102,Priya Verma,ECE,Midterm,2026/02/10,EC201,75,100,Active
```

```
S102,Priya
Verma,Electronics,Final,2026/04/25,EC201,82,100,Active
S103,Rohan Gupta,CSE,Midterm,2026/02/10,CS201,50,100,Active
S103,Rohan Gupta,CS,Final,2026/04/25,CS201,60,100,Active
S104,Sneha Patel,IT,Midterm,2026/02/10,IT201,94,100,Active
S104,Sneha Patel,Information
Tech,Final,2026/04/25,IT201,98,100,Active
S106,Ananya Singh,CSE,Midterm,2026/02/10,CS201,90,100,Active
S106,Ananya Singh,Computer
Science,Final,2026/04/25,CS201,96,100,Active
S107,Rahul Nair,ECE,Midterm,2026/02/10,EC201,70,100,Active
S107,Rahul
Nair,Electronics,Final,2026/04/25,EC201,75,100,Active
S108,Kavya Reddy,IT,Midterm,2026/02/10,IT201,95,100,Active
S108,Kavya Reddy,IT,Final,2026/04/25,IT201,100,100,Active
```

File 3: Course_Credit_Master.csv

```
Subject_Code,Subject_Title,Credit_Hours,Passing_Marks
CS101,Data Structures,4,40
CS201,Algorithms,4,40
EC101,Digital Logic,3,40
EC201,Signals & Systems,4,40
IT101,Database Systems,4,40
IT201,Cloud Computing,4,40
ME101,Thermodynamics,3,40
```

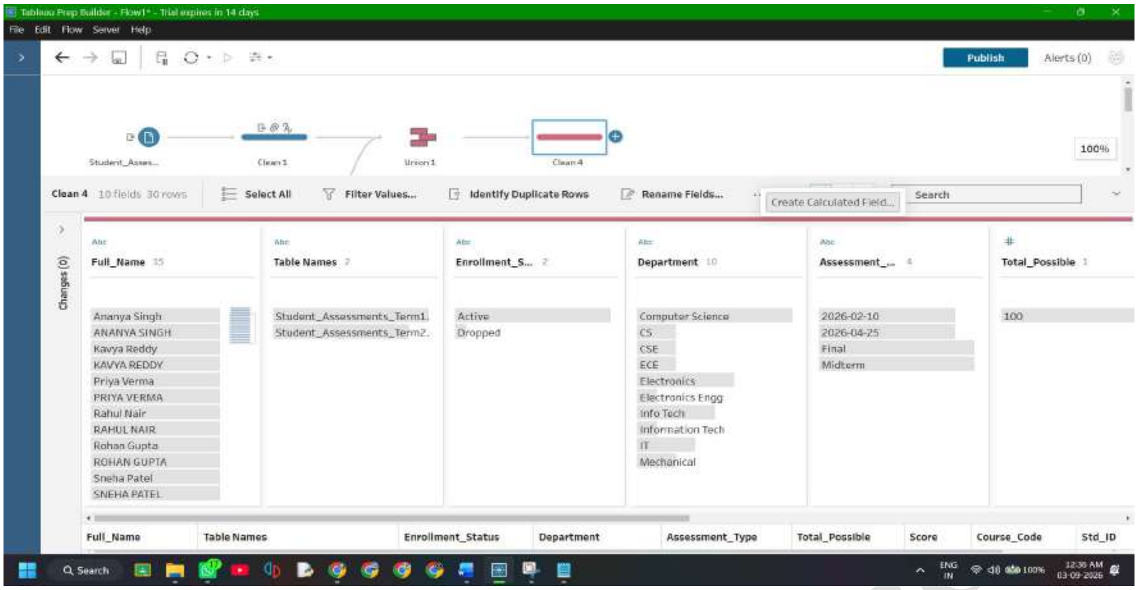
E: Perform Aggregations in Prep

Objective: Create an **Aggregate Step** in Tableau Prep to summarize granular, exam-level test scores into student-level cumulative metrics (Total Marks, Average Score, and Total Assessment Count) per term.

Raw assessment tables log every individual examination (Midterm, Final) on a separate row. By performing an aggregation step directly in Tableau Prep, we roll up individual row events into pre-computed summary statistics, reducing data redundancy and generating student scorecard baselines prior to dashboarding.

Step 1: Add a Term Identifier Field in `Clean 3`

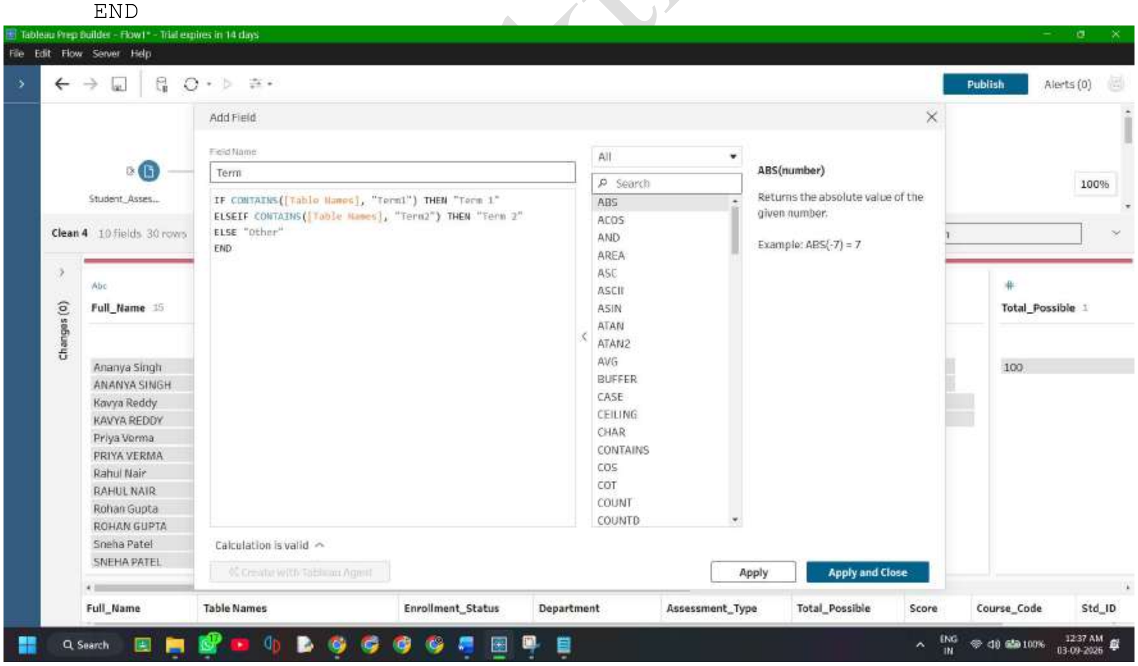
1. Click on the **`Clean 3`** step (the clean step created after `Union 1`).
2. In the toolbar above the Profile cards, click **Create Calculated Field** ***You have to click three dots on the toolbar***.



3. Set the **Field Name** to: `Term`
4. Enter the formula:

   Code snippet

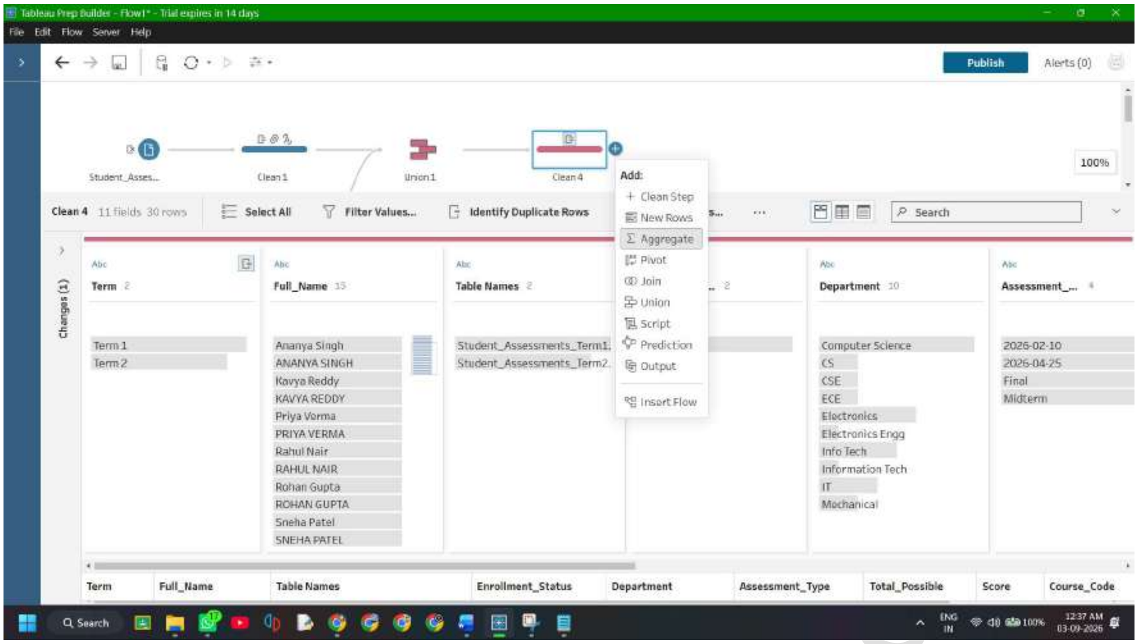
```
IF CONTAINS([Table Names], "Term1") THEN "Term 1"
ELSEIF CONTAINS([Table Names], "Term2") THEN "Term 2"
ELSE "Other"
END
```



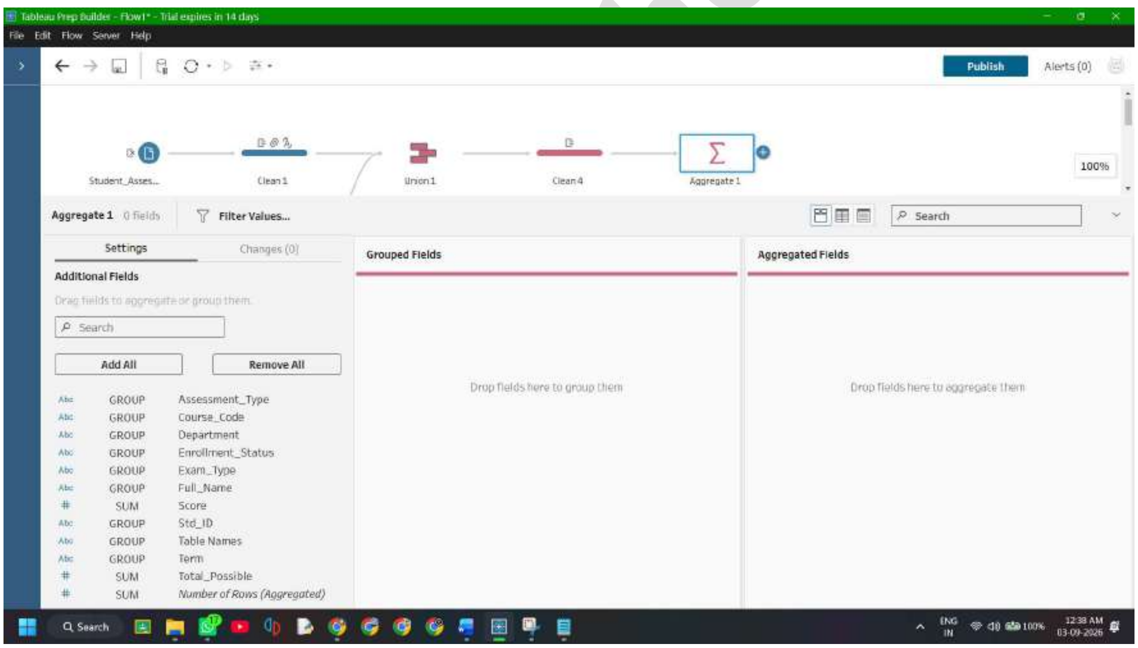
5. Click **Save**.

Step 2: Add an Aggregate Step to the Flow Canvas

1. Hover over the **`Clean 3`** node on the Flow canvas.
2. Click the **+** icon next to it.
3. Select **Aggregate** (represented by the summation icon $\sum$).



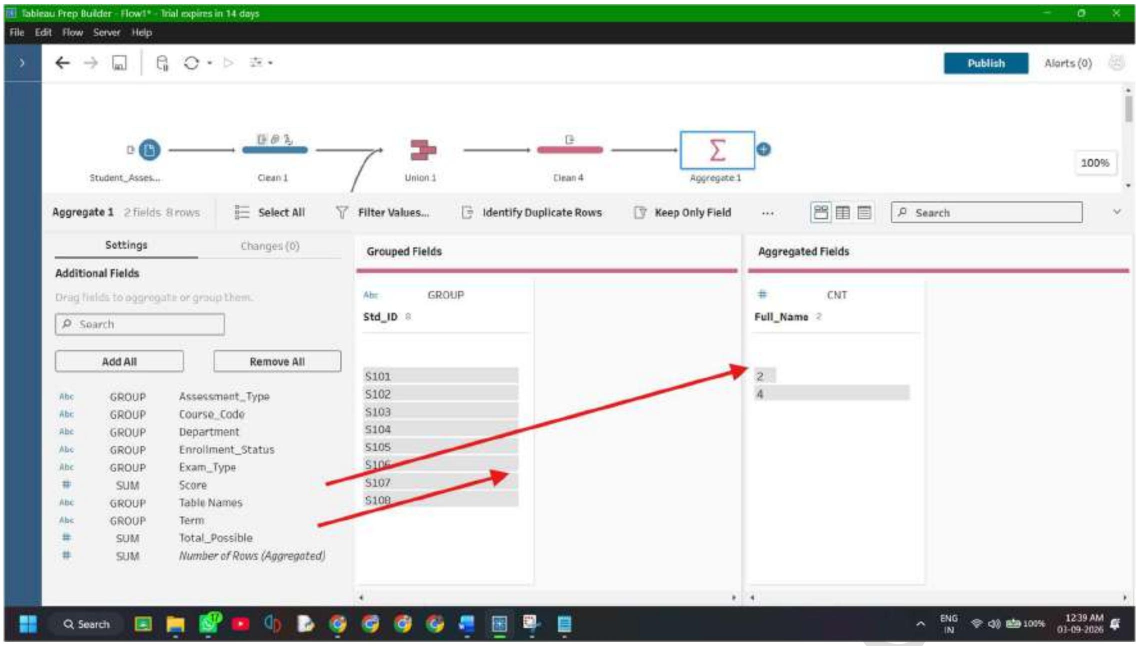
4. An **`Aggregate 1`** step will appear, showing two drop zones in the configuration pane:
   - **Grouped Fields** (Left Zone)
   - **Aggregated Fields** (Right Zone)

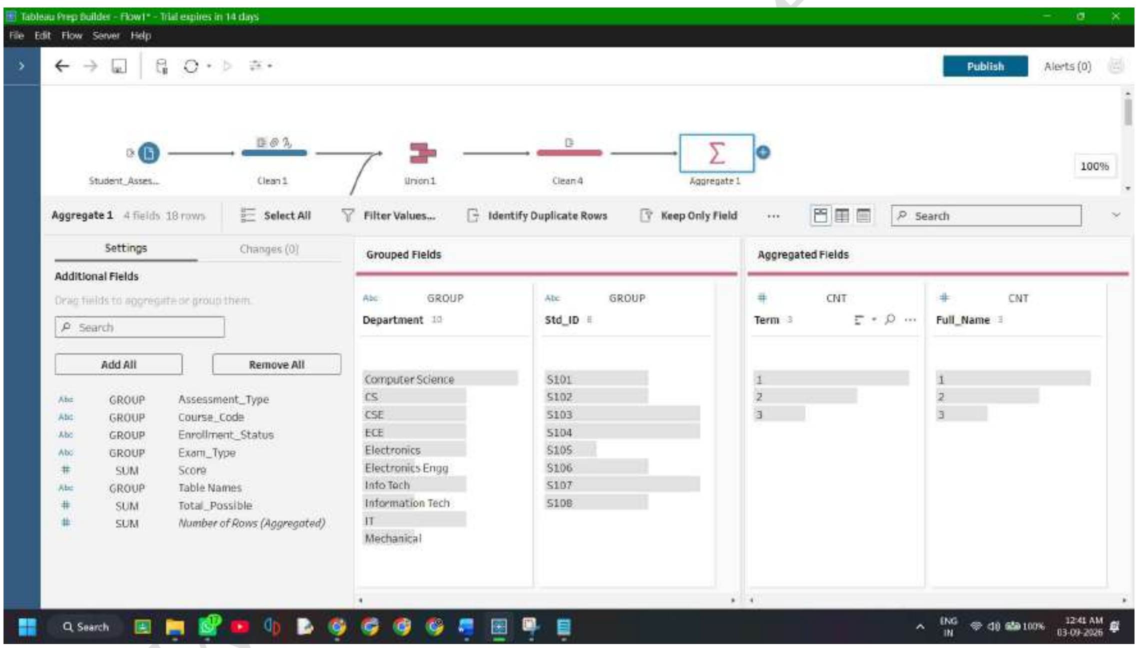


Step 3: Configure Grouped Fields (Dimensions)

From the left-hand column list in the Aggregate step pane *Your Column names could be different based on how you did previous steps of Union*:

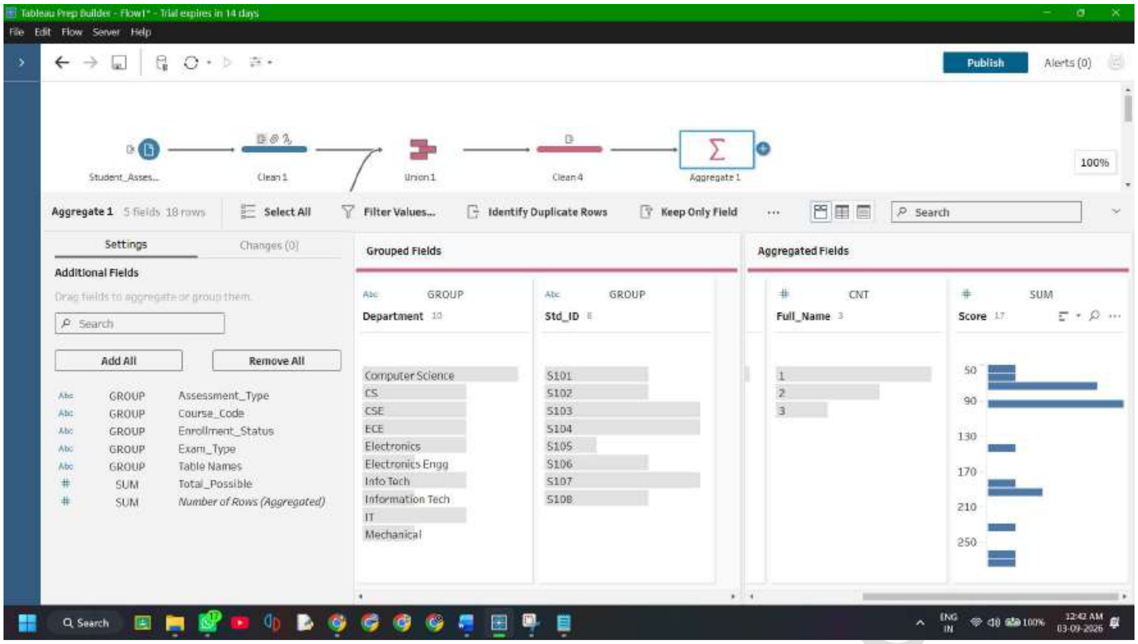
1. Drag **`Student_ID`** into the **Grouped Fields** box.
2. Drag **`Student_Name`** into the **Grouped Fields** box.
3. Drag **`Department`** into the **Grouped Fields** box.
4. Drag **`Term`** into the **Grouped Fields** box.



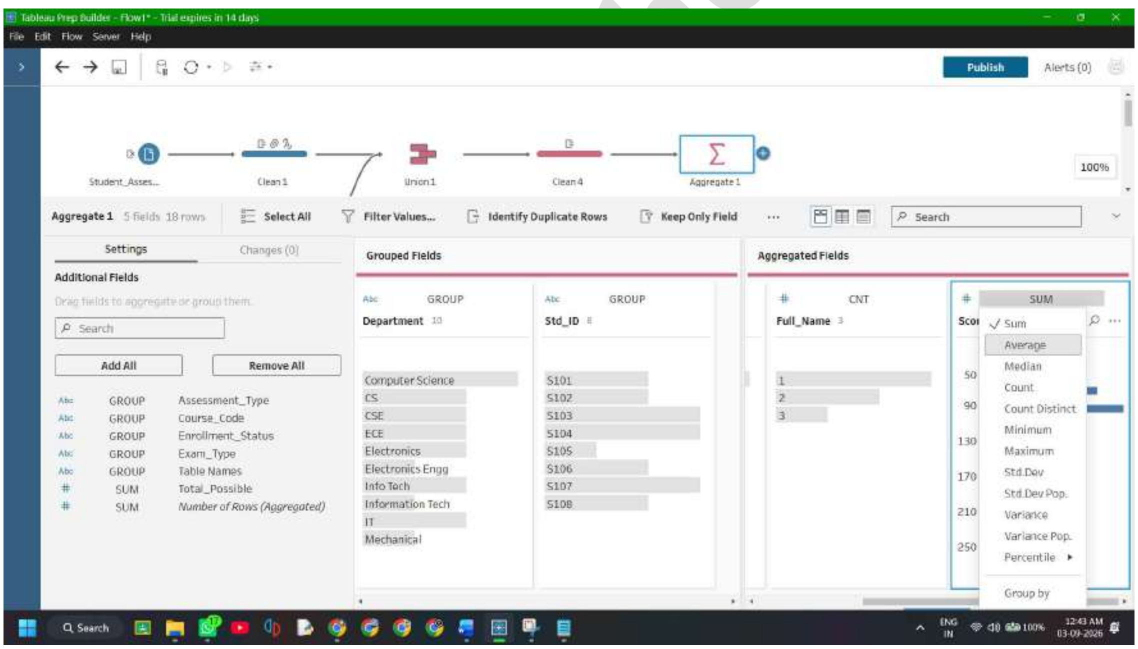


Step 4: Configure Aggregated Fields (Measures & Metrics)

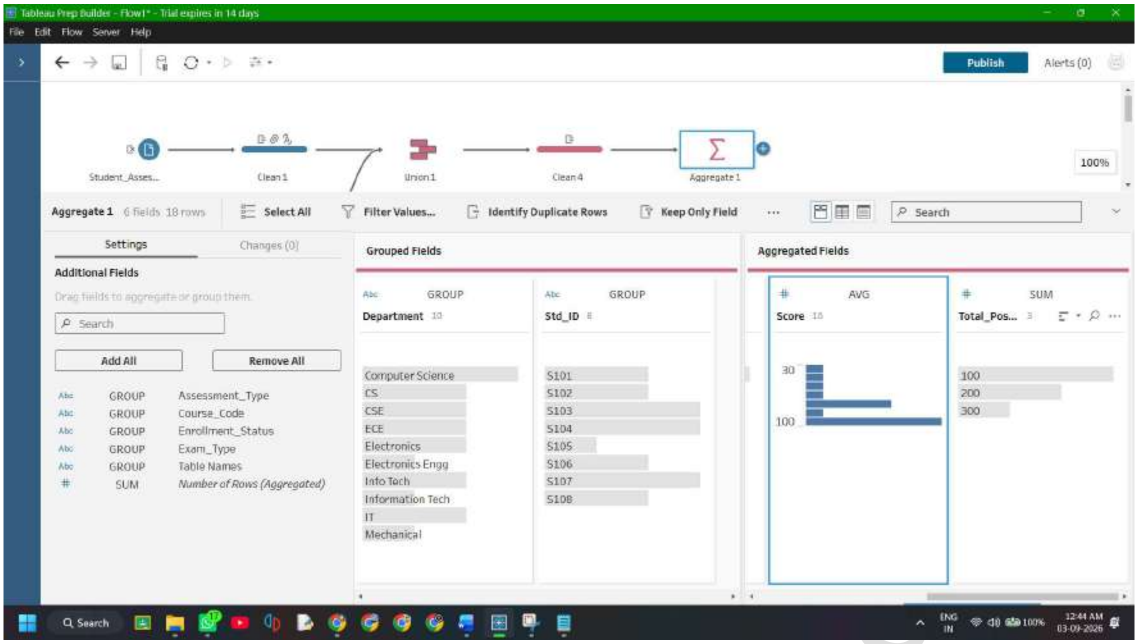
1. **Total Marks:** Drag `Marks_Obtained` into the **Aggregated Fields** box. (Ensure aggregation operation is set to `Sum`).



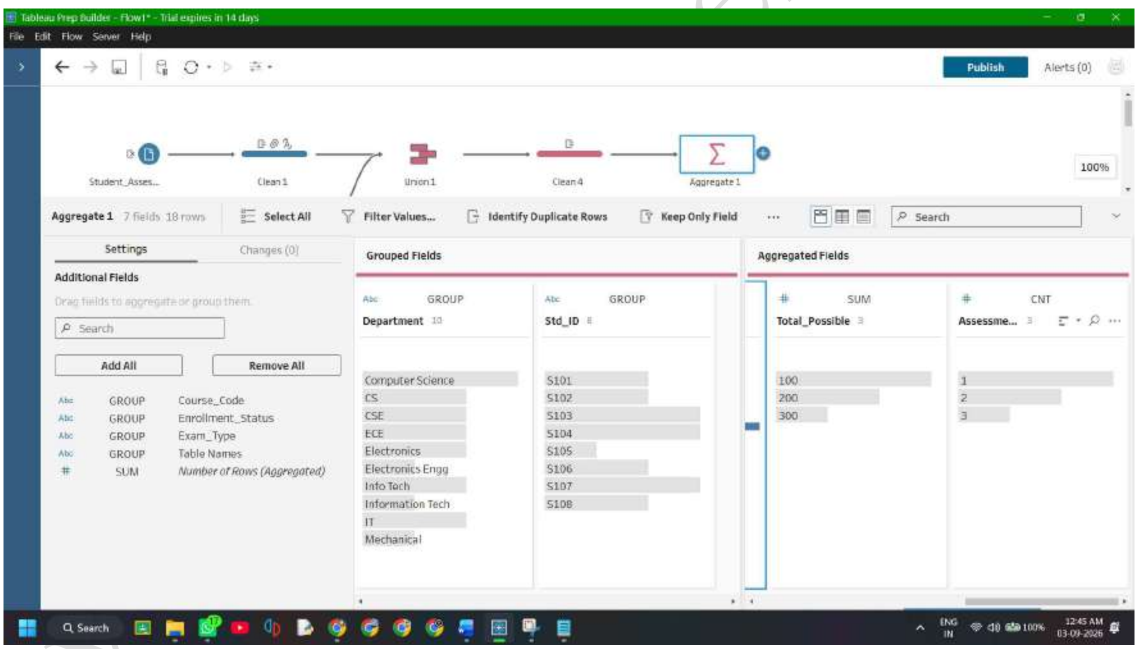
2. **Average Score:** Drag **`Marks_Obtained`** into the **Aggregated Fields** box a second time --> click on `Sum` --> change it to **`Average`**. *There will be no field for dragging by any chance just change SUM to AVERAGE of the existing one.*



3. **Total Possible Marks:** Drag **`Max_Marks`** into the **Aggregated Fields** box --> keep as **`Sum`**.



4. **Assessment Count:** Drag **`Assessment_Type`** (or `Subject_Code`) into the **Aggregated Fields** box --> set aggregation to **`Count`** (or `Count Distinct`).



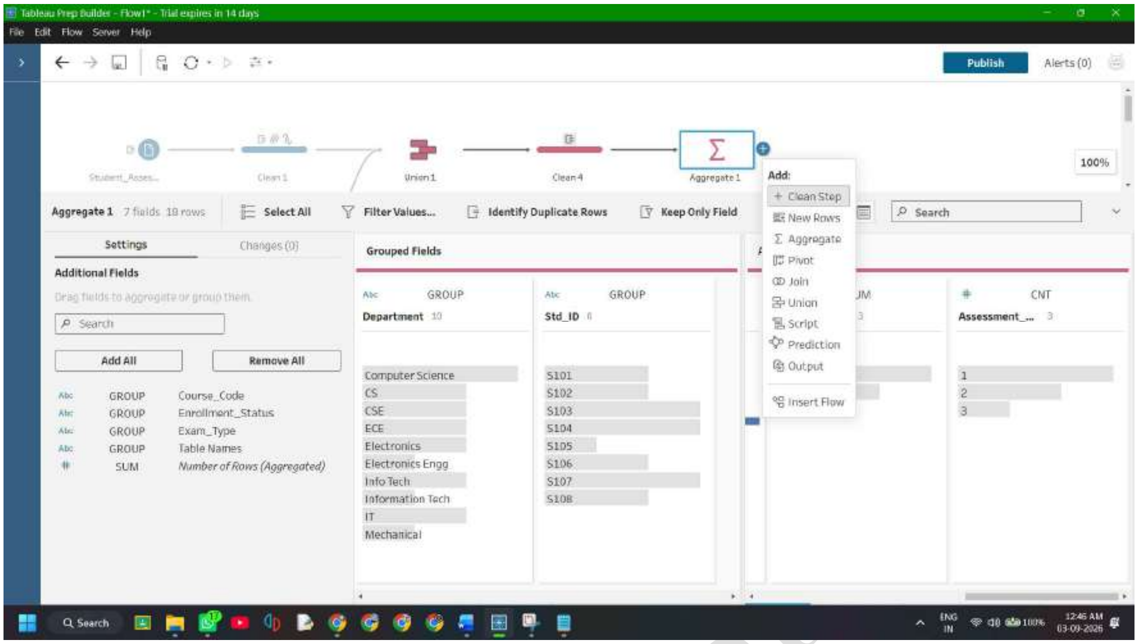
Step 5: Rename Aggregated Output Fields

In the **Aggregated Fields** panel, click on each field name to rename them cleanly:

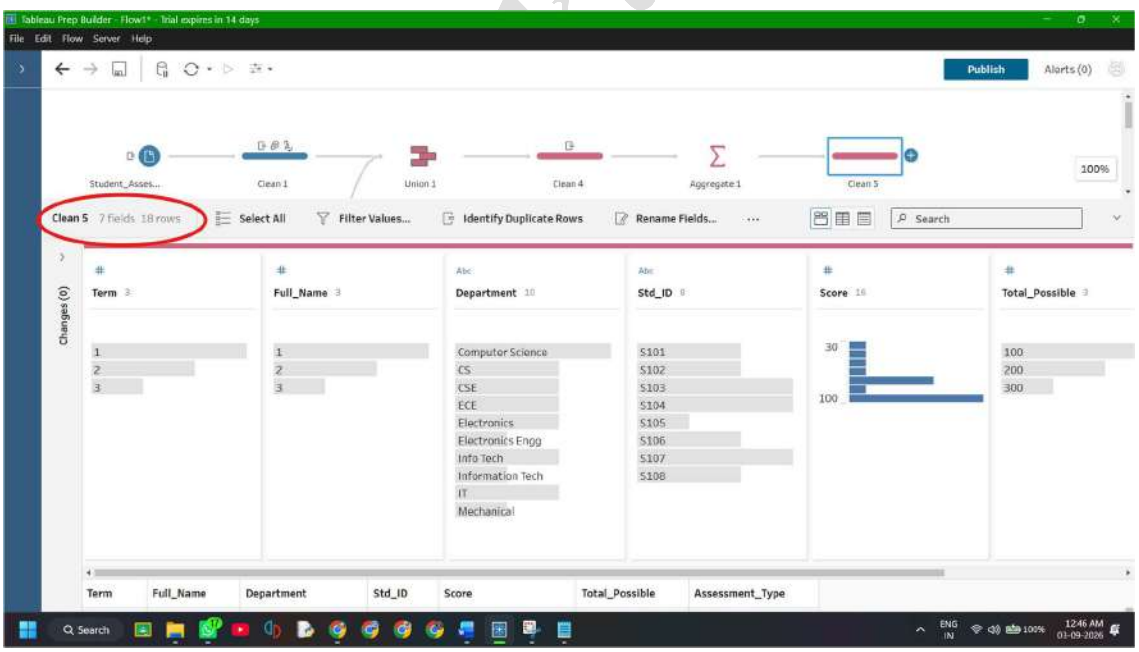
- `Marks_Obtained (Sum)` --> **`Total_Marks_Obtained`**
- `Marks_Obtained (Average)` --> **`Average_Score`**
- `Max_Marks (Sum)` --> **`Total_Max_Marks`**
- `Assessment_Type (Count)` --> **`Assessments_Taken`**

Step 6: Add a Clean Step to Inspect Aggregated Output

1. Click the **+** icon next to the `Aggregate 1` node --> select **Clean Step** (creates `Clean 4`).



2. Look at the Profile Pane:
   - The row count collapses from 30 individual assessment rows into **15 summary records or more** (one row per student per term).



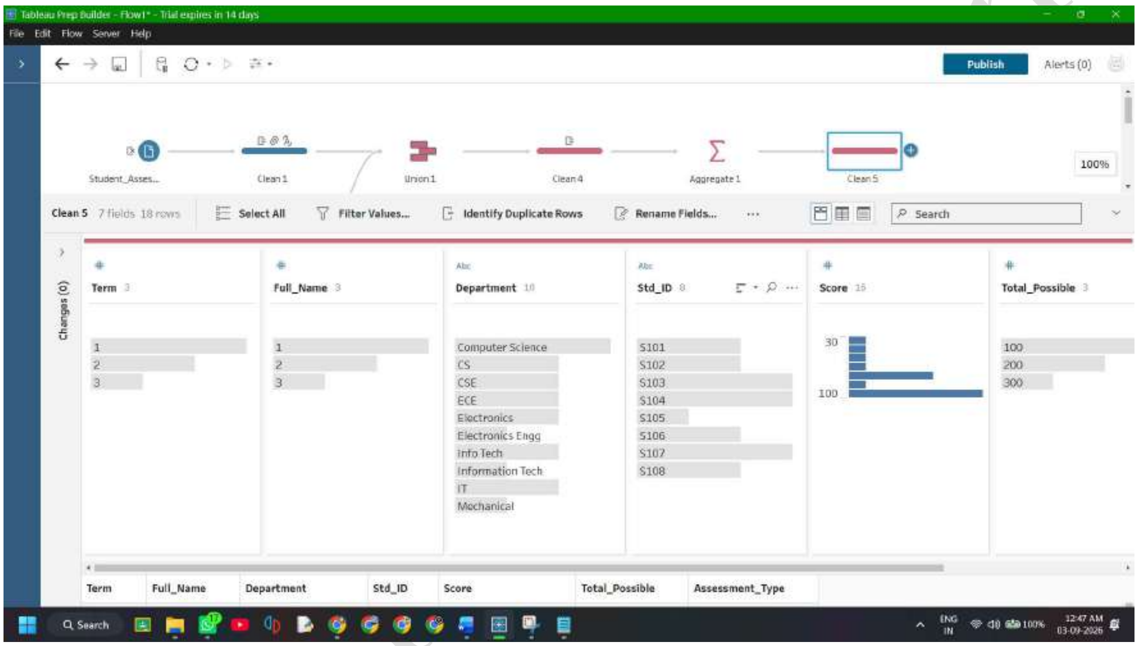
F: Apply Filters and Transformations

Objective: Remove inactive/dropped records, compute weighted percentages, assign automated grading tiers, and derive academic standing indicators using calculated fields in Tableau Prep.

Not all raw records should enter downstream analytical dashboards (e.g., students who dropped out mid-term skew department averages). Transforming raw totals into percentage metrics and standardized letter grades directly inside the data prep pipeline ensures that business logic is applied once and remains consistent across all downstream reporting tools.

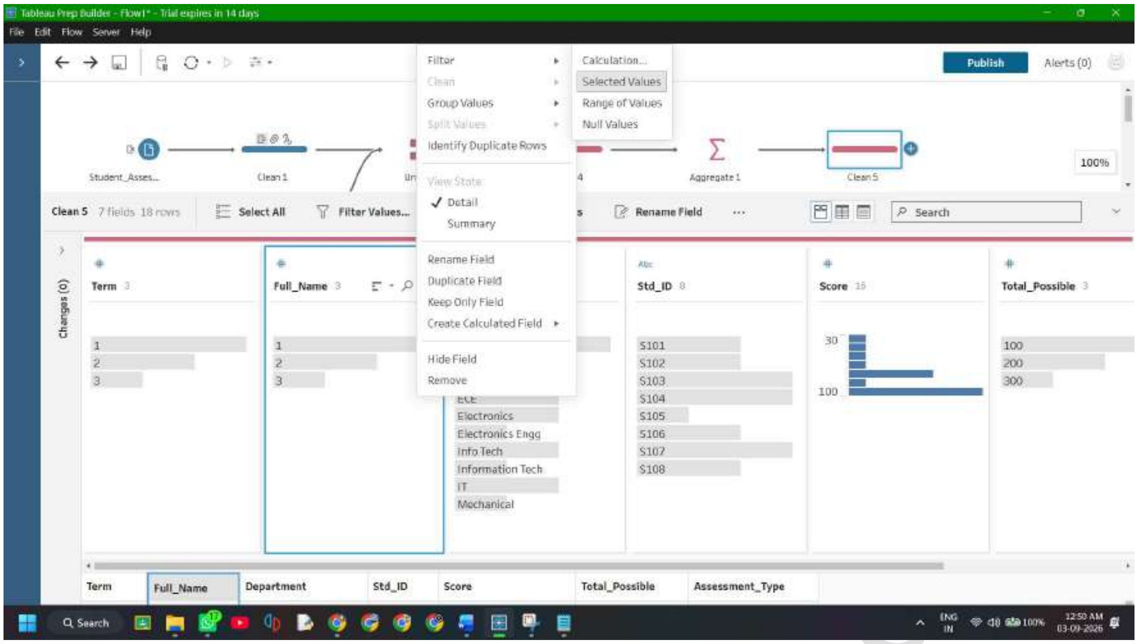
Step 1: Open the Transformation Step (`Clean 4`)

1. Click on the **`Clean 4`** step connected after your `Aggregate 1` step on the Flow canvas.
2. The Profile Pane below will show your aggregated columns: `Student_ID`, `Student_Name`, `Department`, `Term`, `Total_Marks_Obtained`, `Average_Score`, `Total_Max_Marks`, and `Assessments_Taken`. (Naming Will be different and completely depends on how you performed the Union Step)

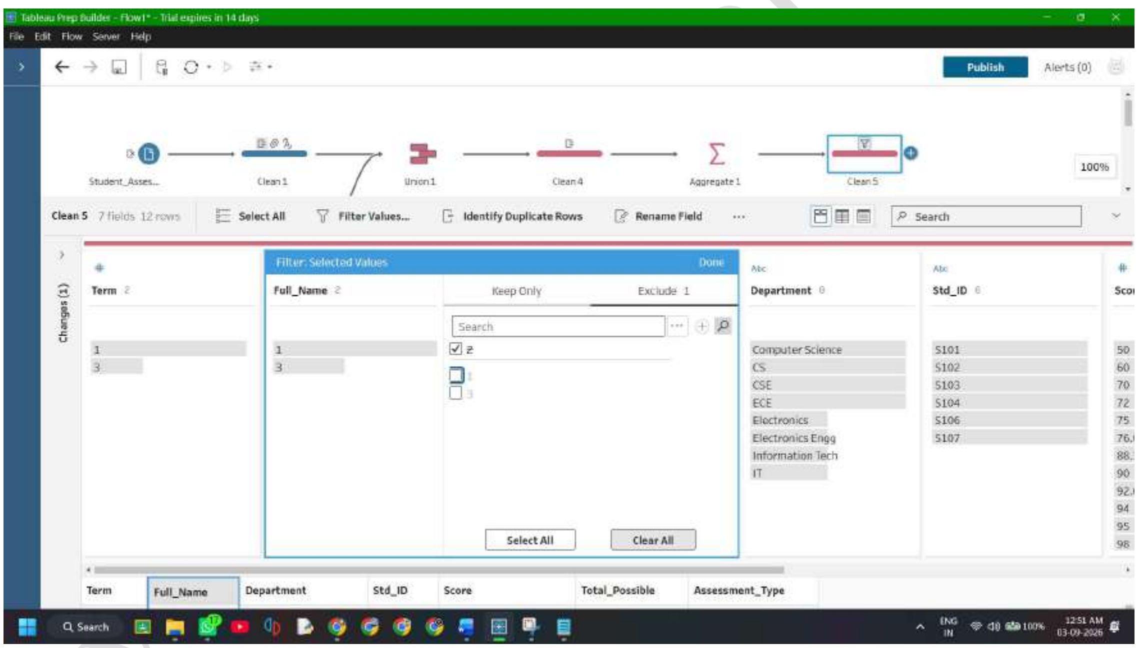


Step 2: Filter Out Inactive / Dropped Student Records

1. If `Enrollment_Status` is present on a profile card:
   - Click on the card --> click the value **`Dropped`** --> right-click and select **Exclude** (or click the filter icon at the top of the card and check only **`Active`**).
2. *(Alternative via Filter calculation)*:
   - In the top toolbar, click **Filter Values** --> **Selected Values...**.



- Exclude `Dropped` so only `Active` students remain in the active pipeline.



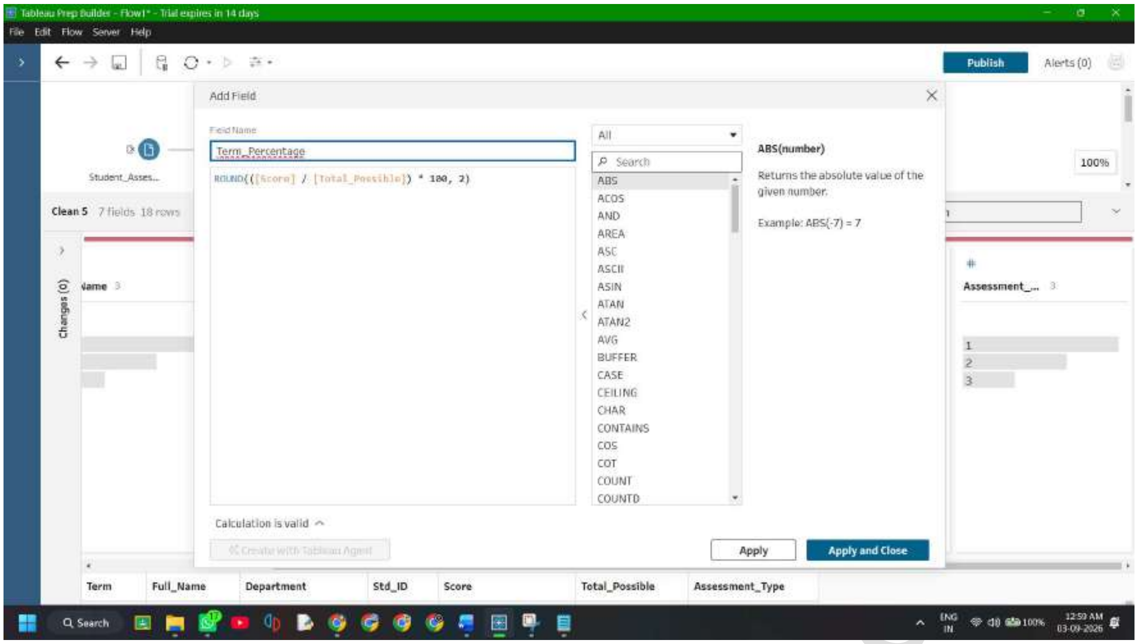
Step 3: Create the Normalized Term Percentage Metric

1. In the toolbar above the Profile cards, click **Create Calculated Field**.
2. **Field Name:** `Term_Percentage`
3. **Tableau Prep Formula:**

   Code snippet

   ```
   ROUND(([Total_Marks_Obtained] / [Total_Max_Marks]) * 100, 2)
   ROUND(([Score] / [Total_Possible]) * 100, 2) {This formula can be
   used if you couldn't able to rename yor aggregate fields.}
   ```

4. Click **Save**.



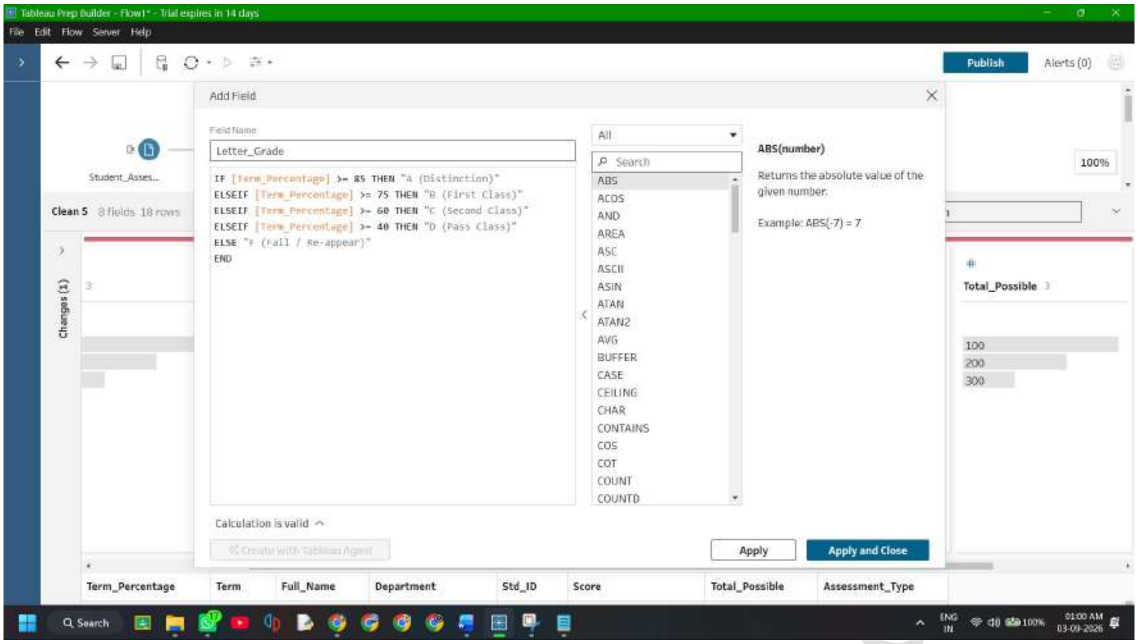
Step 4: Create the Categorical Letter Grade Transformation

1. In the toolbar, click **Create Calculated Field**.
2. **Field Name:** `Letter_Grade`
3. **Tableau Prep Formula:**

   Code snippet

```
IF [Term_Percentage] >= 85 THEN "A (Distinction)"
ELSEIF [Term_Percentage] >= 75 THEN "B (First Class)"
ELSEIF [Term_Percentage] >= 60 THEN "C (Second Class)"
ELSEIF [Term_Percentage] >= 40 THEN "D (Pass Class)"
ELSE "F (Fail / Re-appear)"
END
```

4. Click **Save**.

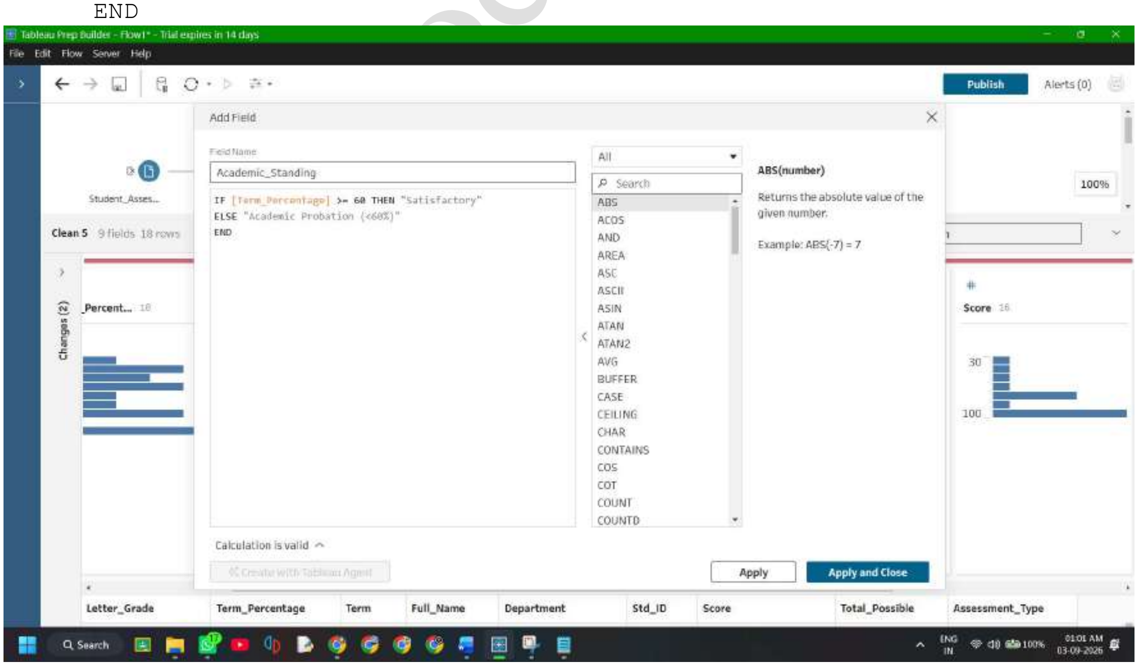


Step 5: Create Academic Standing Compliance Flag

1. Click **Create Calculated Field**.
2. **Field Name:** `Academic_Standing`
3. **Tableau Prep Formula:**

   Code snippet

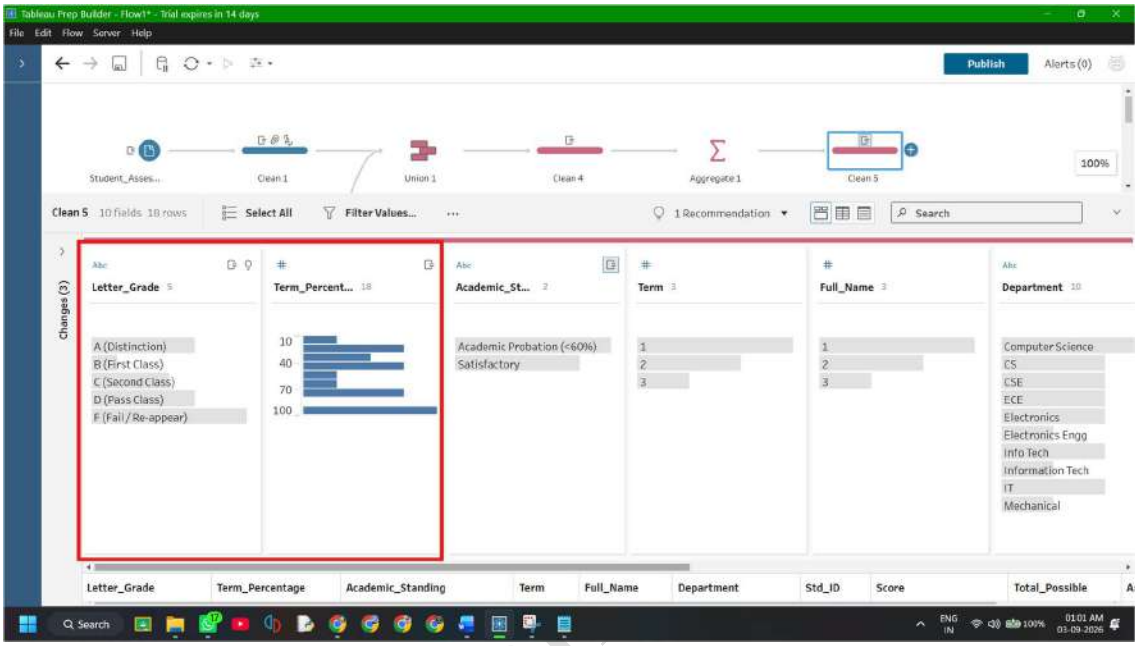
```
IF [Term_Percentage] >= 60 THEN "Satisfactory"
ELSE "Academic Probation (<60%)"
END
```



4. Click **Save**.

Step 6: Verify Transformed Profile Distributions

1. Look at the newly generated profile cards in `Clean 4`:
    - **`Term_Percentage`**: Displays a smooth numeric distribution between 0% and 100%.
    - **`Letter_Grade`**: Displays discrete categorical buckets (`A (Distinction)`, `B (First Class)`, etc.).
    - **`Academic_Standing`**: Highlights students requiring academic counseling.



Transformation Summary Table

Transformation	**Type**	**Formula / Rule**	**Output Meaning**
Filter Status	Row Exclusion	`[Enrollment_Status] != 'Dropped'`	Removes dropped students
`Term_Percentage`	Numeric Calculation	`([Total_Marks_Obtained] / [Total_Max_Marks]) * 100`	Normalized 0–100 scale
`Letter_Grade`	Conditional Dimension	`IF [Term_Percentage] >= 85 THEN "A (Distinction)" ...`	Standardized letter grade
`Academic_Standing`	Conditional Flag	`IF [Term_Percentage] >= 60 THEN "Satisfactory" ...`	Compliance warning tag